

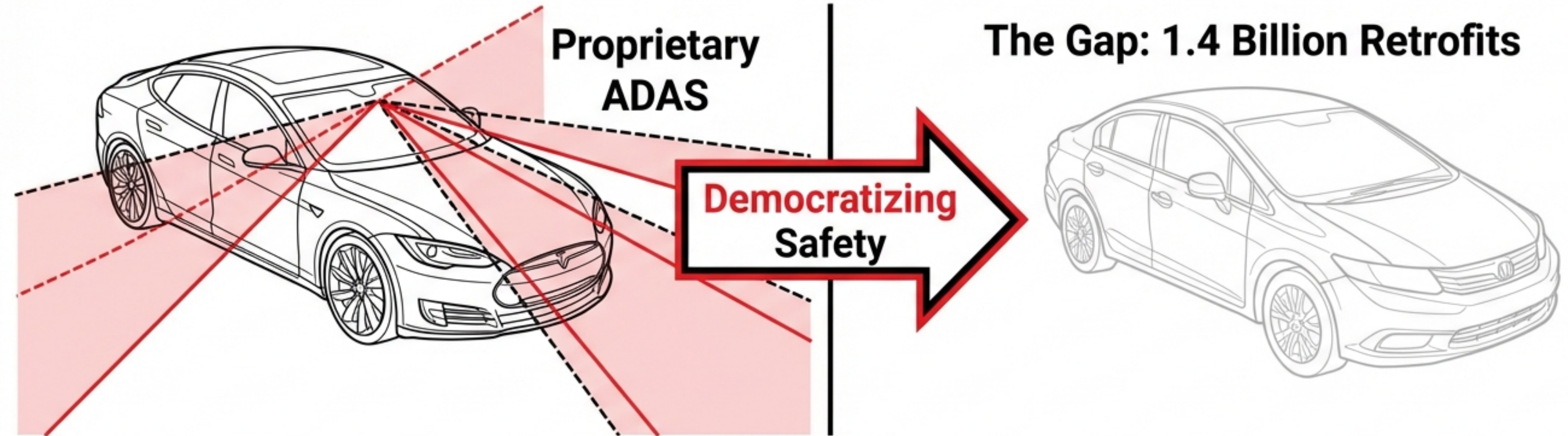


Customizable AI-Enabled ADAS Demonstrator

Capstone Design Project | Group 66

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The Safety Gap: 1.4 Billion 'Blind' Vehicles

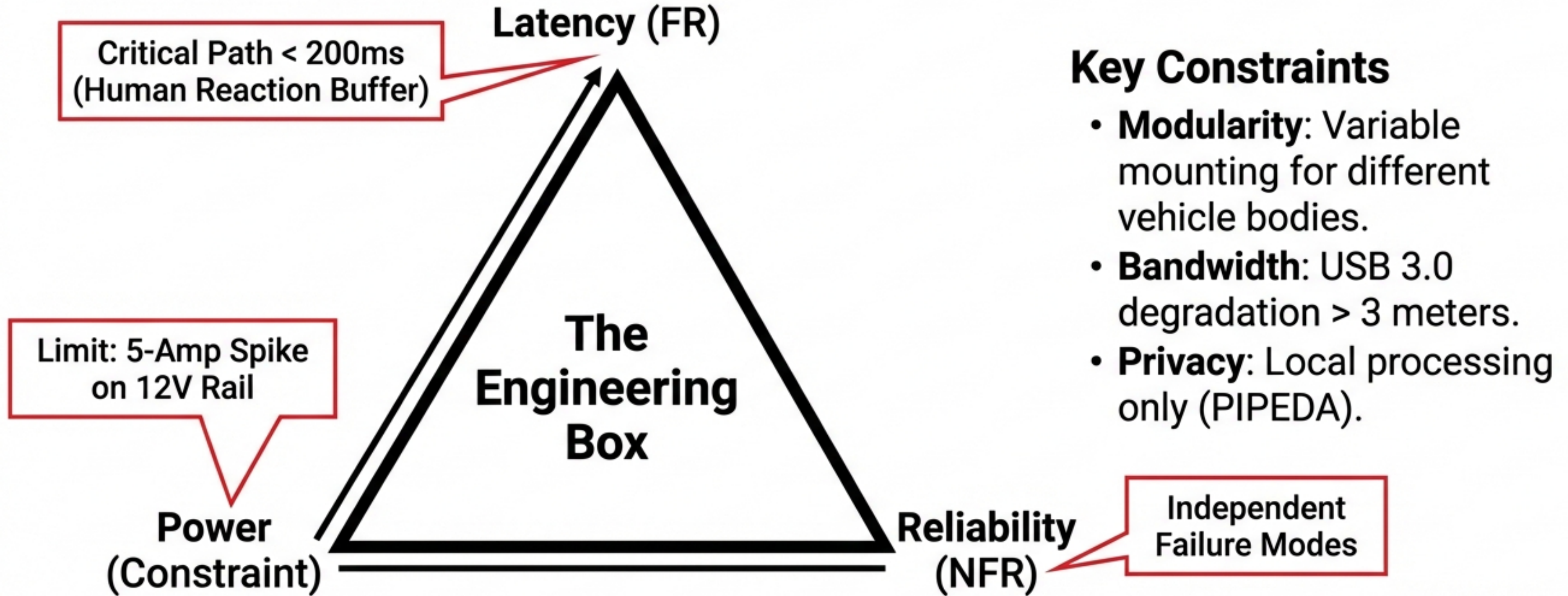


1.19 Million
Annual Fatalities

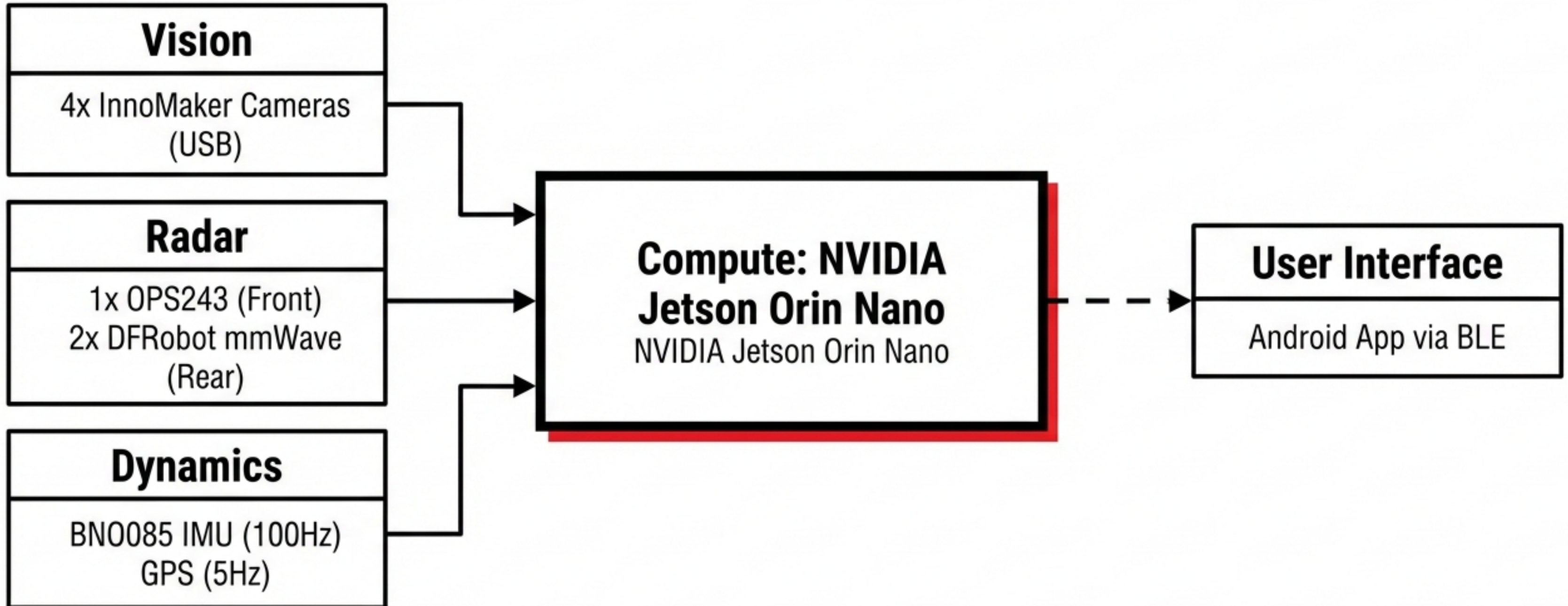
Majority caused by
Perception Failure

Goal: Modular,
Aftermarket Fusion

Engineering Constraints & Requirements



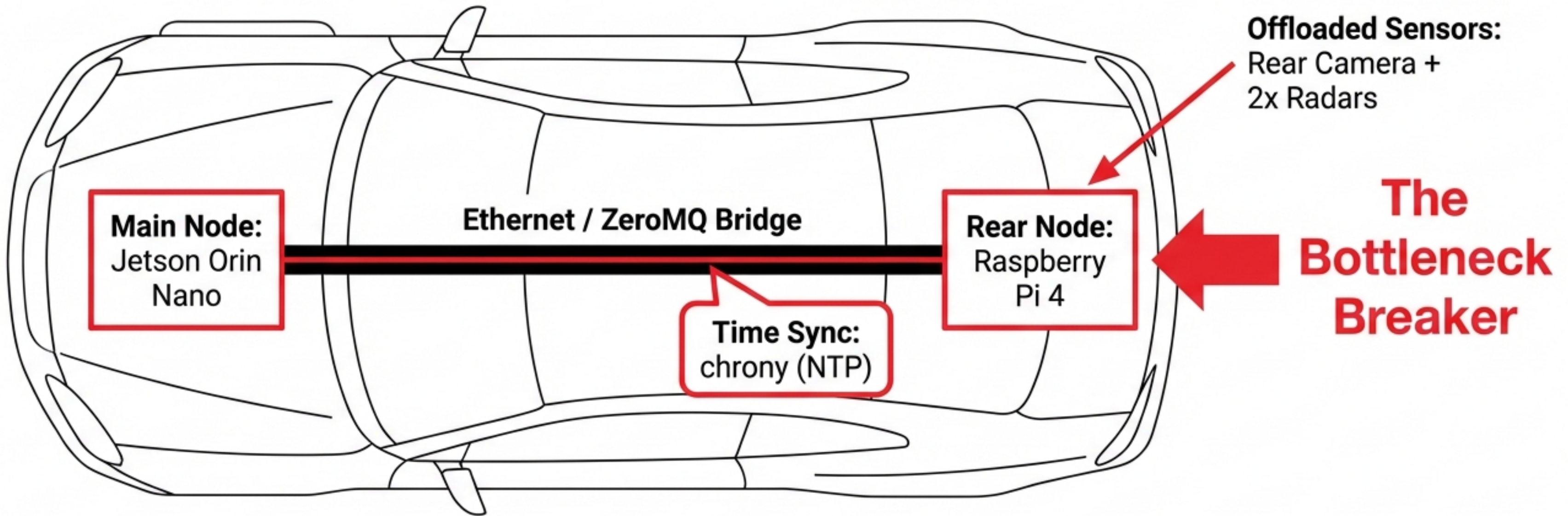
High-Level Hardware Architecture



Design by Exclusion: Alternatives Analysis

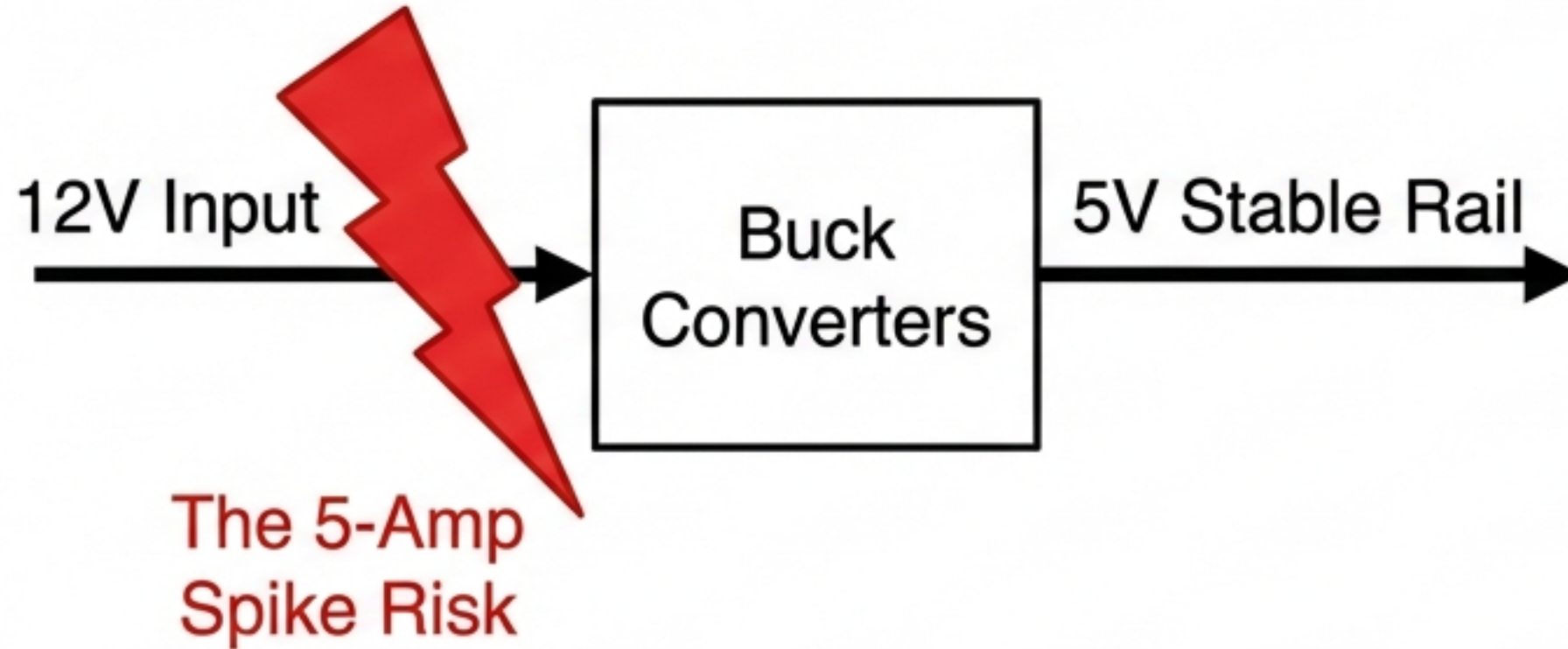
Alternative	Why Not?	Verdict
ROS (Robot Operating System)	High overhead, non-deterministic latency.	REJECTED (Custom C++ Chosen)
LiDAR (RPLidar A1)	5.5Hz scan rate. At 60km/h, target moves 3m per scan.	REJECTED (Too Slow)
Centralized USB Architecture	4x Cameras saturate USB bandwidth. 3m cables signal loss.	CRITICAL FAILURE (Pivot Required)

The Pivot: Distributed Hardware Architecture



Power Engineering & Signal Integrity

Custom Power Distribution Board

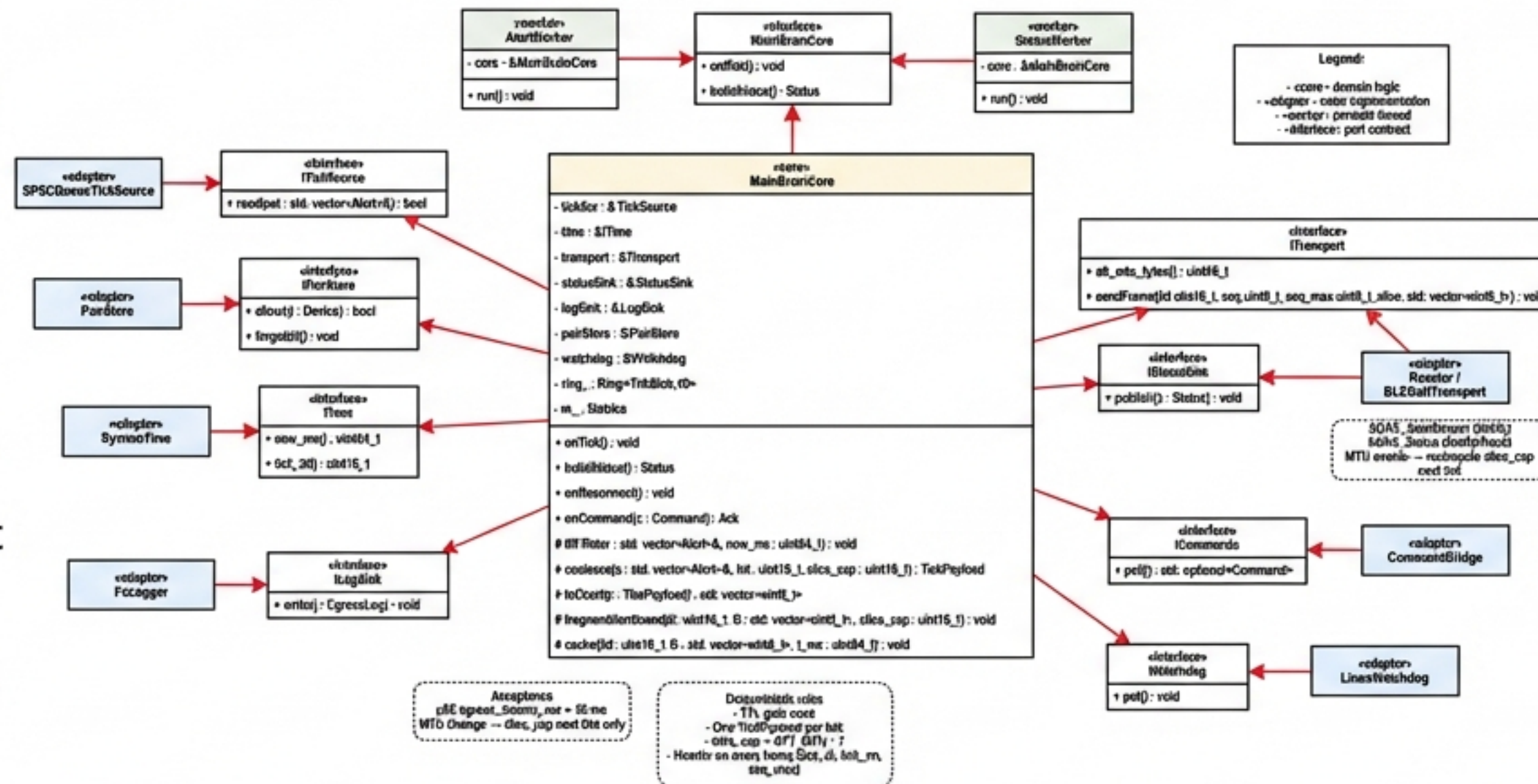


- Problem: Inrush current $> 5A$ trips car adapters.
- Solution: Step-down regulation (12V $\rightarrow$ 5V).
- Safety: Dedicated fusing per NFRs.
- Signal Integrity: Ethernet bridging for 3m runs.

The Perception Engine: C++ Pipeline

Philosophy: Freshness over Completeness

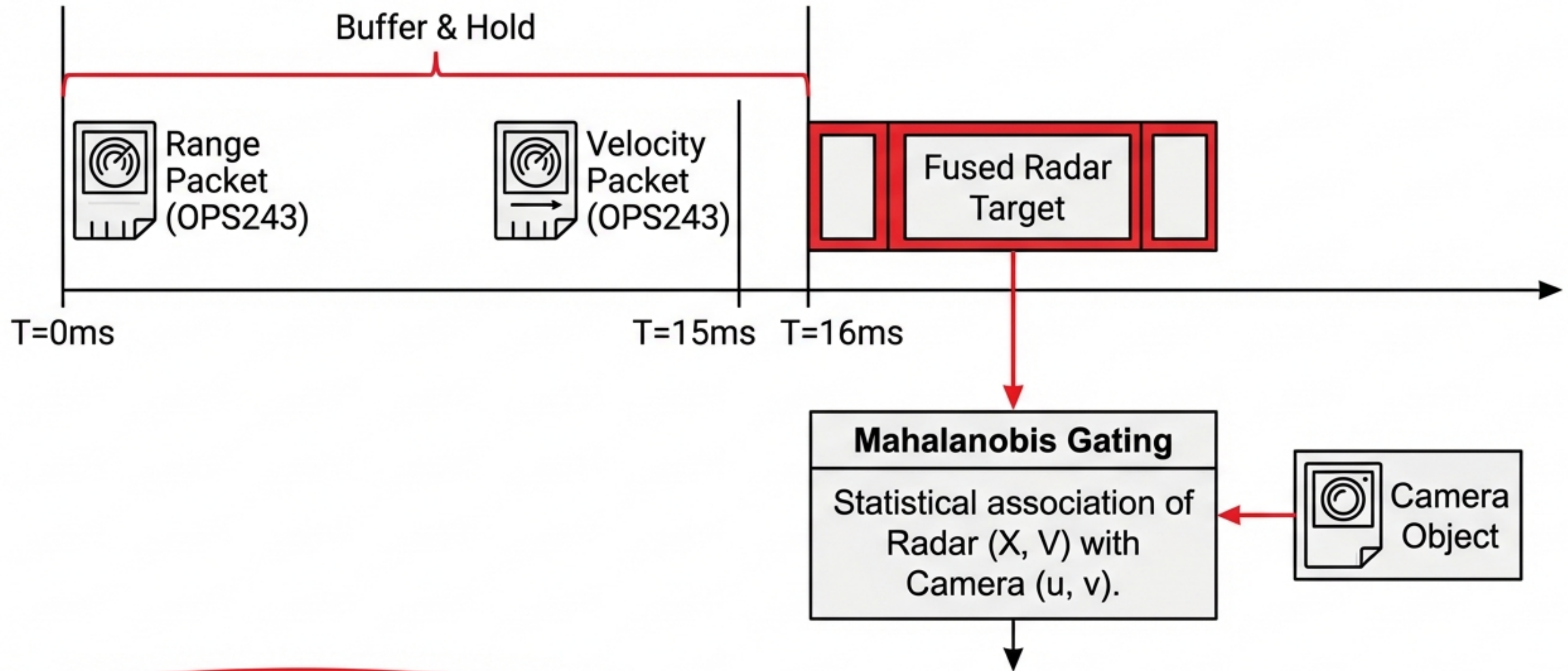
Freshness over completeness have accomplished brace frames created and being determinies. So, make understand, to that on demand then would on over complement ratios.



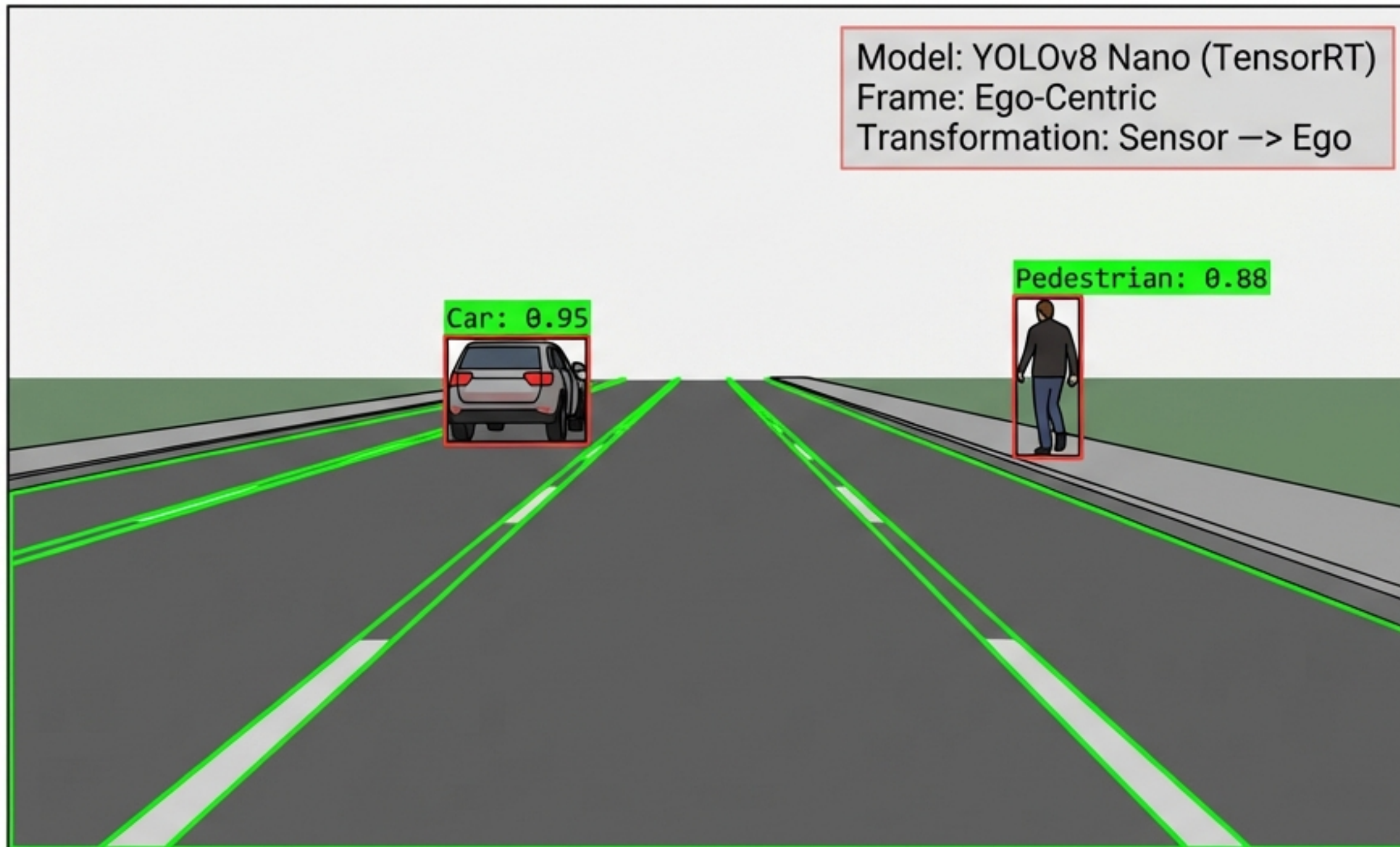
Mechanism:

- Lock-free SPSC Queues.
- Old frames dropped to preserve 20Hz Tick.
- Migrated Python -> C++ for deterministic timing.

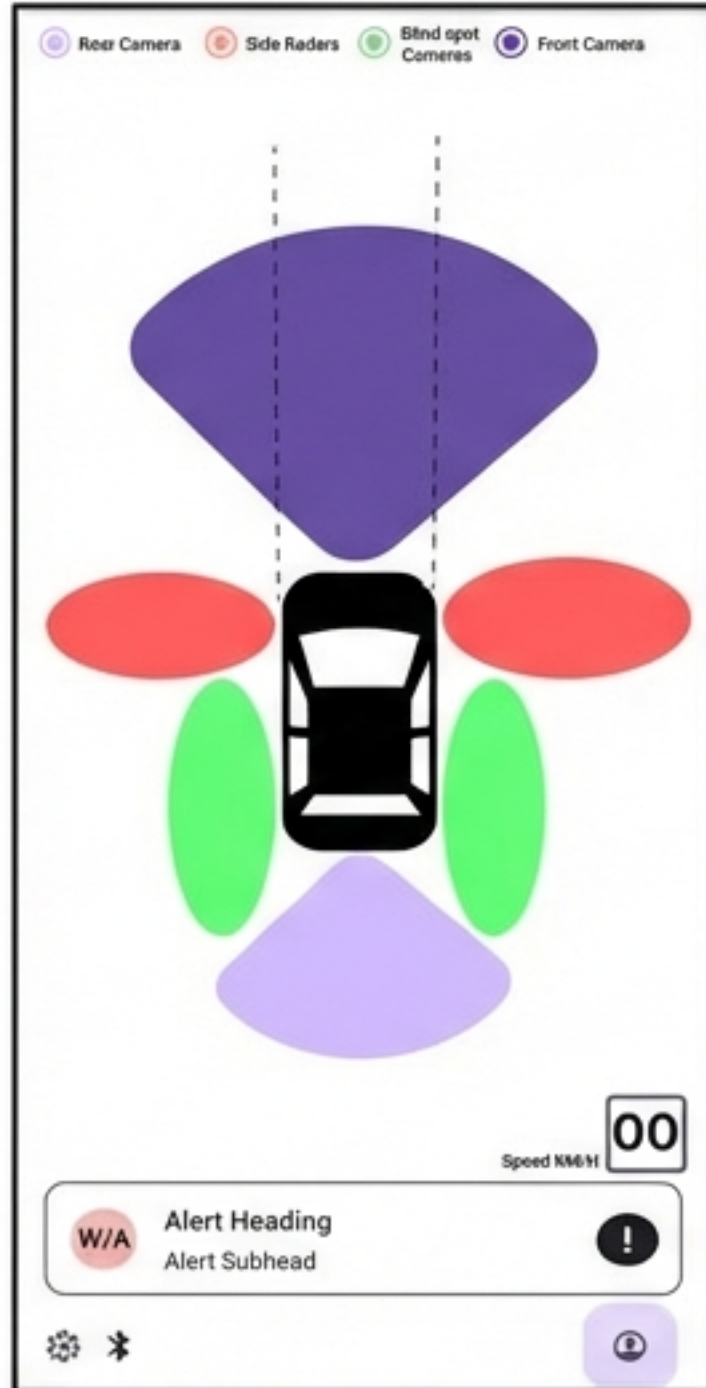
Sensor Fusion: The 'Hold & Fuse' Algorithm



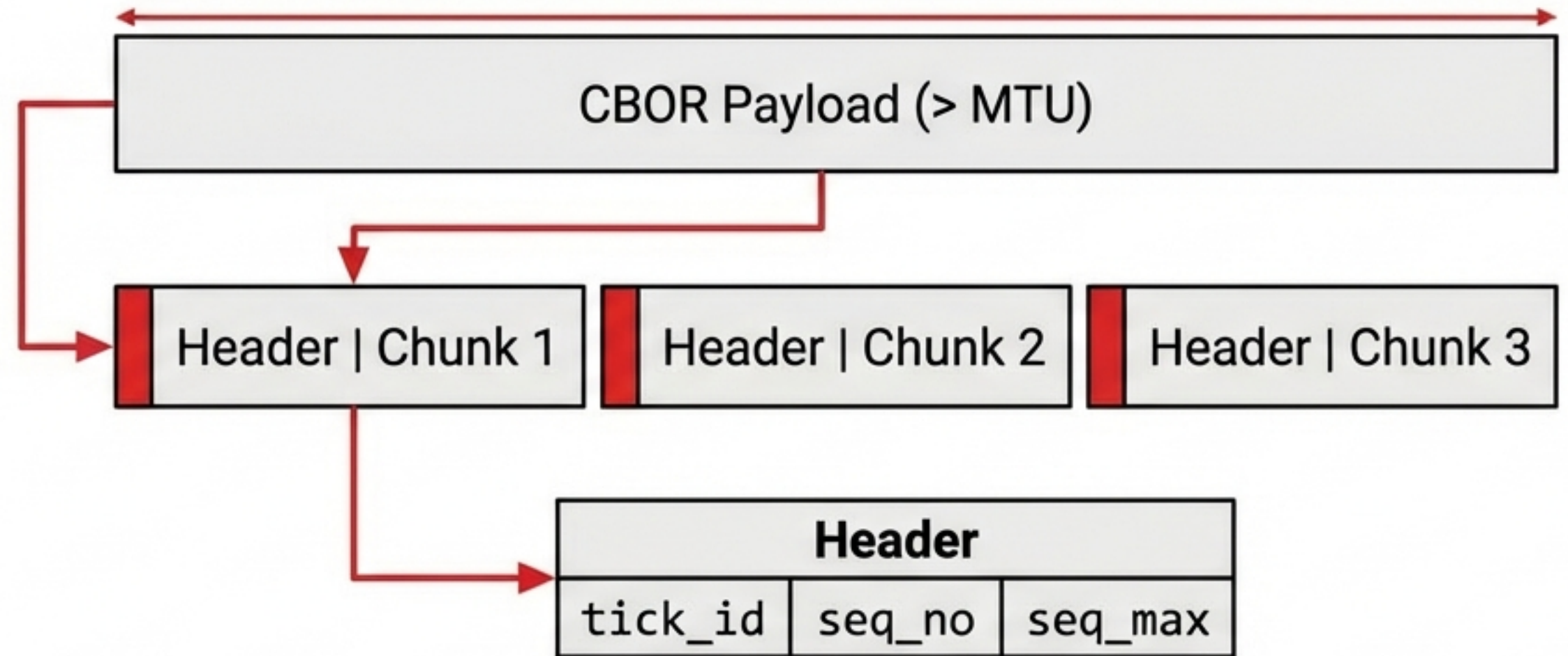
Computer Vision & Lane Perception



Mobile Architecture & The BLE Link



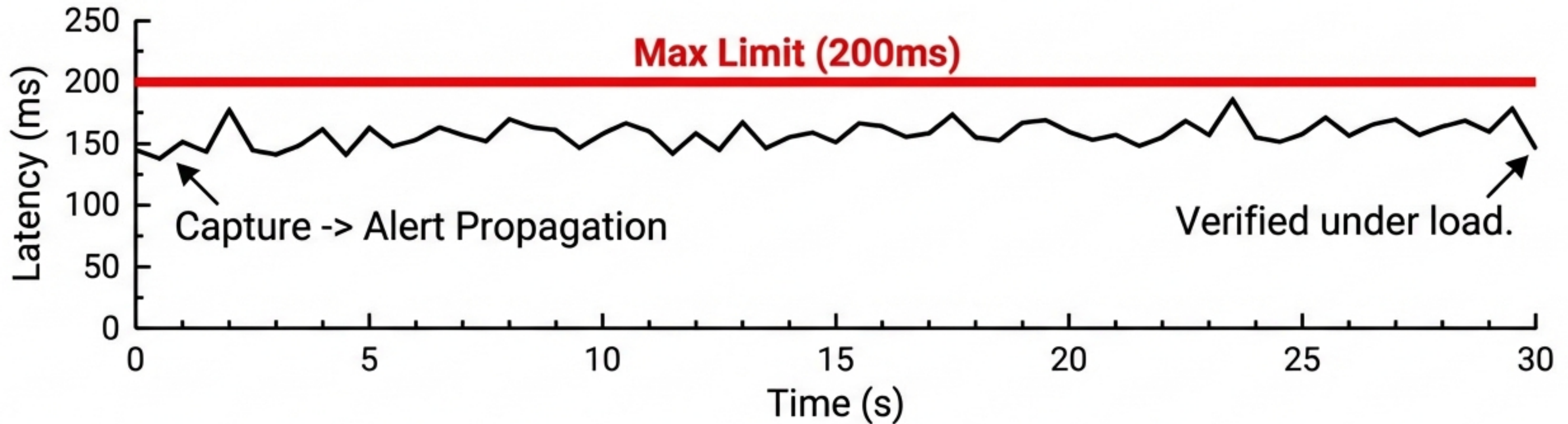
BLE Fragmentation Strategy



Manual Fragmentation & Reassembly for high-bandwidth alerts.

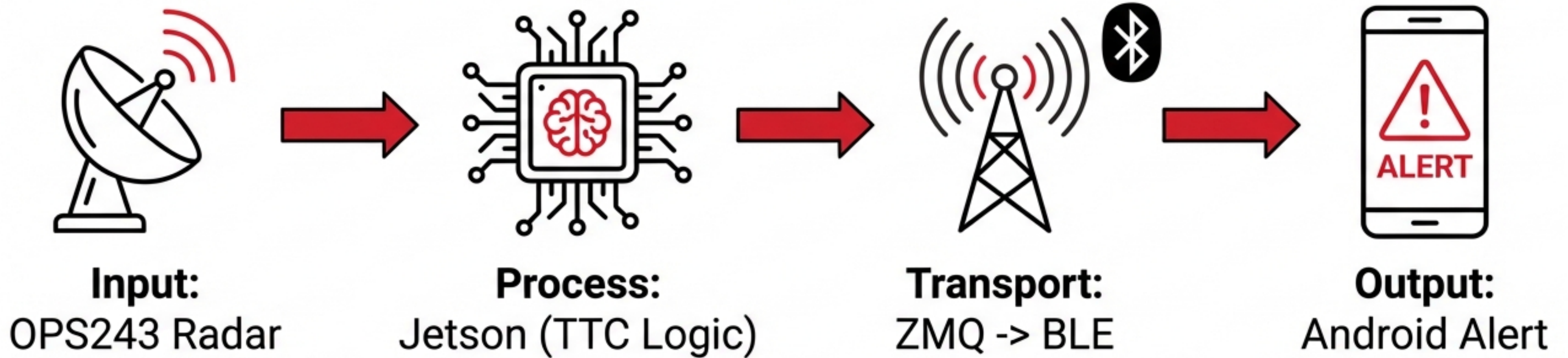
Verification: Validating the Critical Path

System Latency Over Time



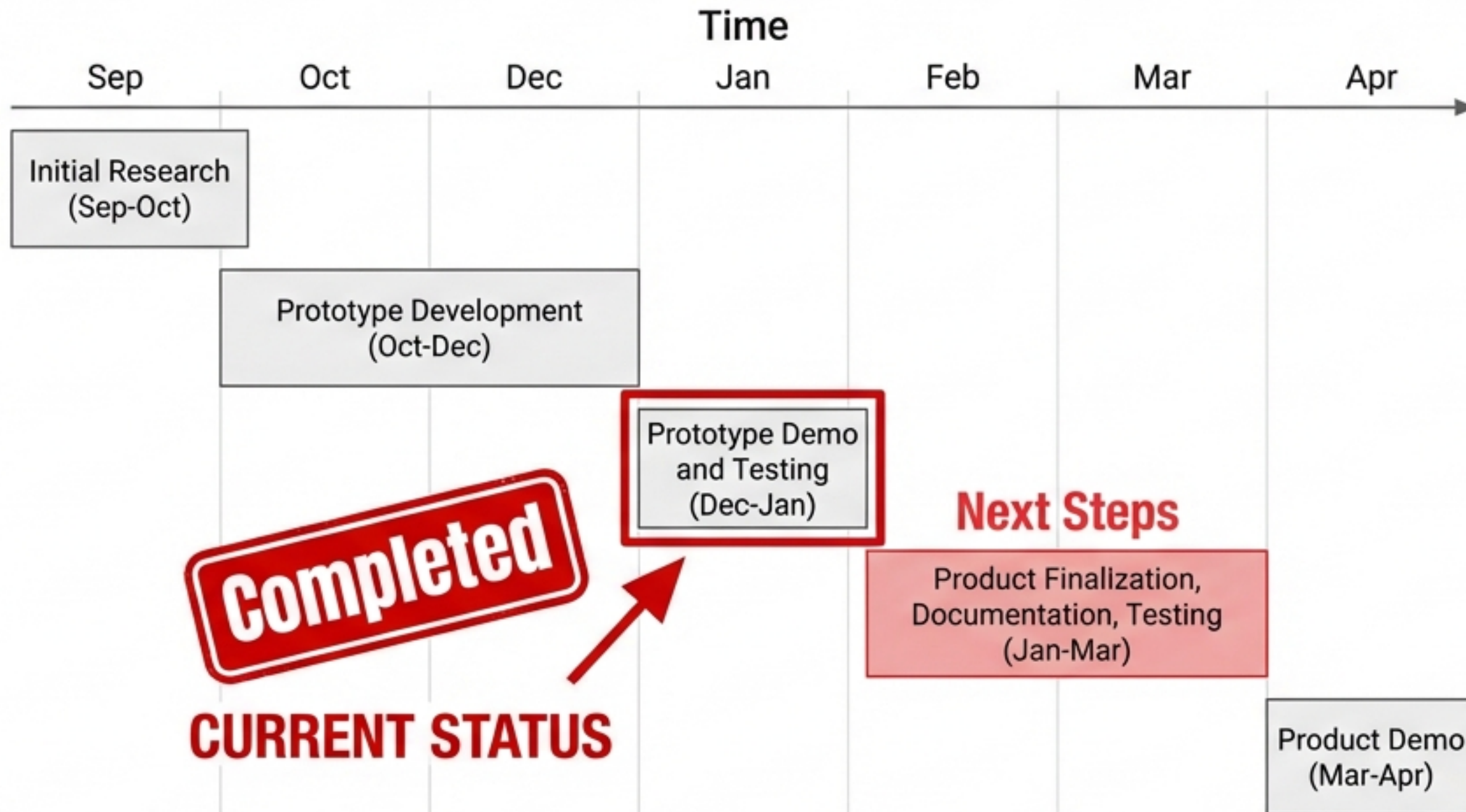
- **Unit Tests:** SPSC Queue overflow behavior confirmed.
- **Integration:** Deterministic Replay (FR81) passed.

The Vertical Slice: Proof of Concept



Significance: Validates electrical & logical critical path before full sensor integration.

Project Roadmap & Status



Next Steps

1. YOLOv8 Vision Model Integration.
2. Field Testing on Honda Civic.
3. Refining “Ghost Box” Fusion visualization.

Summary of Engineering Achievements



- **Architecture:** Distributed Edge Computing (Jetson + Pi).
- **Performance:** <200ms Latency via C++ Pipeline.
- **Safety:** Robust Power & Privacy-First Design.